NAFSI

NAFSI HUB DATA CHALLENGE FALL 2026, Sens-D track

Proposed solution

PoultryRescue: Salmonella small ground



Lot-level decisions for poultry plants

Submitted by: Team

Early Signal

Information,

expertise and contributions of group members

Blessing Ademola: PhD student, Chemical and Biomedical Engineering, University of Missouri. Research background in precision agriculture sensing, GIS and spatiotemporal data analysis.

For the NAFSI Challenge, I contributed to project conceptualization, data sourcing, data integration, and forecast integration.

Friederike Albrecht: I am a PhD student at the University of Missouri Sinclair School of Nursing (MUSSON), where my research focuses on nutrition, food security, and health disparities among under-resourced populations. My background in nursing and doctoral-level research provides a foundation in epidemiology, pathophysiology, microbiology, disease prevention, and population health. My research experience includes community-based participatory research (CBPR) and

community–academic partnerships, including the *From the Ground Up* study, which supports low-income households through home gardening, nutrition education, and health screenings.

For the NAFSI Challenge, I contributed to the conceptualization and initial development of PoultryRescue, integrating food safety, regulatory considerations, and food security. By examining the USDA-FSIS 2024 regulatory framework, I identified important distinctions between poultry product categories, contamination thresholds, and opportunities for safe alternative processing, informing the development of a product-specific decision tree. Using Kiro, I developed the initial Python framework incorporating *Salmonella* risk-response pathways, product classification, alternative processing decisions, and identification of potential processing facilities. I also integrated food security and health equity considerations, emphasizing the preservation of safe, nutritious poultry, reducing avoidable food loss, and improving access to dietary protein in under-resourced communities.

Javonte Carter: I am a Master of Science student in Computer Science at Florida Agricultural and Mechanical University (FAMU), where I also earned my Bachelor of Science in Information Technology with a focus on Health Informatics and Information Assurance and Security. My academic and research interests include artificial intelligence, machine learning, cybersecurity, high-performance computing, and blockchain technology.
My current graduate research focuses on CertChain, a blockchain-based micro-credential framework designed to automate academic credential eligibility and verification in higher education. This work integrates natural language processing, machine learning, cryptographic security, and distributed ledger technology to improve the reliability, transparency, and efficiency of digital credentialing. I have also participated in high-performance computing research initiatives, including HackHPC, the SGX3 Coding Institute, and the Gateways 2026 conference, where I presented research on data-grounded generative AI using Python, structured datasets, and large language models. These experiences strengthened my skills in software development, AI integration, data processing, and the evaluation of computational systems.
For the NAFSI Challenge, I brought a computing and software perspective to the team, took part in discussions on the project direction and the design of the decision-support tool, and reviewed the application from the perspective of a user who needs clear, actionable recommendations.

Nafisa: I am a PhD student in Food and Hospitality Systems at the University of Missouri–Columbia and a graduate research assistant at the FEAST Lab. My academic and research expertise includes food science, nutrition, food security, food processing, and sustainable food systems. Before pursuing my doctoral studies, I worked for approximately five years in humanitarian nutrition and food security programs with UNICEF, BRAC International, and Concern Worldwide, supporting vulnerable populations in the Rohingya refugee response in Bangladesh. This experience provided me with a strong foundation in nutritional assessment, food security, and interdisciplinary collaboration.

For the NAFSI Challenge, I contributed my food science and nutrition expertise to the development and refinement of the team's decision-support framework, particularly in reviewing food-safety considerations, nutritional outcomes, and the relationship between food safety and food security. I also participated in reviewing the analytical workflow, identifying areas for improvement, and supporting the preparation of the final report. Additionally, I initiated efforts to

engage food science specialists from the University of Missouri to seek expert perspectives and facilitate interdisciplinary discussion among team members. My contributions focused on strengthening the scientific relevance, practical applicability, and communication of the proposed solution.

Waqar: I am a PhD student in Civil and Environmental Engineering at the University of Missouri, advised by Dr. Binbin Wang. My research combines hydrologic and hydraulic modeling, machine learning and geospatial data for flood forecasting, including a graph neural network flood model and a digital twin of Hinkson Creek in Columbia, Missouri. That work gave me experience in building data pipelines, forecasting and testing models against held-out data, which shaped how this project was evaluated.
For the NAFSI Challenge, I led the data analysis and software. I assembled and linked the FSIS sampling, category posting and MPI directory data, designed and ran the published-category forward test, the rebuilt-status check and the test without the 2025 posting, and brought Blessing's state forecast and Nafisa's safety review into one decision engine. I built the PoultryRescue web app and the public GitHub repository with its tests and data dictionary, ran the Kiro work, and produced the report figures. I used AI coding assistants for much of the code and checked the results against the source data.

Team Science Statement

Team Early Signal formed organically when five individuals who were interested in the NAFSI HUB Data Challenge connected to participate. As work began, our complementary backgrounds in data analytics, computer science, engineering, food science, and nursing became important strengths. The range of expertise helped the team consider both how to build a functioning analytical tool and how its outputs might inform real-world food-safety and food-security decisions.
The computational and engineering perspectives supported analysis of public datasets, forecasting, software development, and implementation of the decision-support framework. Food science expertise added attention to processing practices, microbial controls, and the nutritional consequences of food loss. The nursing and population-health perspective linked potential Salmonella exposure with prevention, access to nutritious food, and the needs of under-resourced communities. This exchange helped shift the project toward a processing-plant decision-support application that considers product-specific regulatory constraints, practical management alternatives, and food preservation together. Individual contributions are described above; the collaboration demonstrates how an initially informal team can develop an integrated approach through complementary disciplinary strengths.

Contents

Team Science Statement 3
Executive summary 5

Introduction 6
Problem Statement and Rationale 7
Burden and attribution 8
Performance standards and their limits 8
Plant-level evidence 8
Forecasting, season and weather 9
Controls, lots and diversion 9
Food security 10
The gap 10
Data 11
Methods 12
Category forward test 12
State case forecast 13
Decision logic 14
County need 15
Results 15
Published category predicts next-year positivity 15
State forecast 17
Decision scenarios 19
Food security 21
The application 22
Kiro 23
Discussion and Public Health Implications 23
Limitations 24
Reproducibility 25
Use of AI tools 25
Conclusion 25
Responses to the challenge questions 26

Executive summary

Nontyphoidal *Salmonella* remains a leading cause of foodborne illness, hospitalization, and mortality in the United States (Scallan et al., 2011; Walter et al., 2025), with poultry serving as a major source of human exposure (CDC, 2026a; Rose et al., 2025). For small and very small poultry processors, managing *Salmonella* contamination while avoiding unnecessary food loss presents a significant challenge at the intersection of food safety and food security. Emerging quantitative pathogen-detection technologies and evolving regulatory requirements create opportunities for more informed, product-specific processing decisions.

PoultryRescue is a proactive decision-support platform developed to help Quality Assurance (QA) and Hazard Analysis and Critical Control Point (HACCP) professionals translate *Salmonella* surveillance information and available microbiological testing results into appropriate product-management decisions. The framework integrates product-specific regulatory requirements, contamination risk, alternative processing options, and food-preservation considerations. Its intended application includes supporting decisions when quantitative *Salmonella* measurements become available, particularly where regulatory thresholds may influence product disposition.

To establish an early-warning component, the project analyzed four consecutive annual USDA Food Safety and Inspection Service (FSIS) reporting periods. Ground chicken processing lines classified as Categories 1, 2, and 3 subsequently exhibited *Salmonella* positivity rates of 13.4%, 23.3%, and 47.3%, respectively. Combining establishment category with product type improved predictive performance, reducing the Brier score by 4.3% relative to a product-only baseline (95% CI: 2.9–6.0%). A separate three-month state-level surveillance forecasting model outperformed conventional baselines by 5.1–12.1%. However, additional FSIS and environmental variables did not improve forecasting accuracy, and declining alert performance in 2026 led to the use of these forecasts as contextual information rather than direct inputs for operational recommendations.

PoultryRescue translates these findings into an explainable decision framework that compares six potential management responses, including additional verification, product holding, validated further processing, and disposal when necessary. The interactive web-based prototype evaluates modeled exposure reduction, food retention, nutritional protein preservation, and potential distribution to communities experiencing greater food insecurity. Illustrative scenarios demonstrate how eligible poultry products may be preserved through validated cooking while maintaining applicable food-safety requirements.

By connecting early risk indicators and available contamination information with practical processing alternatives, PoultryRescue aims to support timely, regulation-informed decisions while reducing avoidable food loss. Future development will focus on integrating validated quantitative pathogen measurements and operational processing constraints. The platform does not independently determine product safety, authorize product release, predict outbreaks, or estimate illnesses prevented; all consequential decisions remain subject to applicable regulations and review by qualified food-safety professionals.

Introduction

Food safety and food security are interconnected public health priorities that influence nutritional health, overall well-being, and the sustainability of food systems. Among foodborne pathogens, nontyphoidal *Salmonella* continues to represent a major public health concern in the United States. The Centers for Disease Control and Prevention (CDC) estimates that *Salmonella* is responsible for approximately 1.35 million infections each year, most of which are linked to contaminated food products (CDC, 2024a). Poultry products, particularly chicken, provide an important source of high-quality protein and essential nutrients but are also recognized as significant sources of *Salmonella* exposure (CDC, 2025b; Choi et al., 2023). As a result, effective control of contamination throughout poultry production, processing, and distribution remains essential for protecting public health. At the same time, efforts to improve food safety should consider their potential effects on food availability, nutrient retention, and the long-term sustainability of the food supply.

The connection between food safety and food security is especially important for vulnerable populations. According to the USDA Economic Research Service, approximately 13.7% of U.S. households, equivalent to 18.3 million households, experienced food insecurity in 2024 (Rabbitt et al., 2025). Limited access to nutritious food can adversely affect health outcomes, while foodborne infections may pose particularly serious risks for young children, older adults, and immunocompromised individuals (CDC, 2025a). Moreover, food-safety incidents, product recalls, and disruptions in poultry-processing operations can result in the loss of food that might otherwise be recovered and utilized. Previous research on U.S. meat and poultry recalls has demonstrated the relationship between food-safety events and food waste (Gorton & Stasiewicz, 2017). Therefore, approaches that maintain rigorous food-safety protections while preserving safe and nutritious food may contribute to a more resilient, efficient, and equitable food system.

To support pathogen control in poultry processing, the USDA Food Safety and Inspection Service (FSIS) has established product-specific Salmonella performance standards and an establishment categorization system (USDA, 2016). These measures provide insight into the effectiveness of pathogen-control programs at individual processing establishments. However, historical surveillance data alone do not determine the safety of individual poultry lots or identify the most appropriate operational response. Decisions regarding product management must also consider microbiological testing results, processing capabilities, regulatory requirements, and the feasibility of alternative disposition strategies. In 2024, FSIS established an adulteration determination for certain not-ready-to-eat breaded stuffed chicken products containing Salmonella at concentrations of 1 CFU/g or greater (Salmonella Not Ready-To-Eat Breaded Stuffed Chicken Products, 2024). Subsequent delays in associated verification activities due to limitations in available testing methods further highlighted the challenges of translating microbiological findings into practical regulatory and operational actions (FSIS, 2025).

These challenges underscore the need for decision-support systems that translate available Salmonella testing results and surveillance information into practical, product-specific management options. PoultryRescue was developed to support Quality Assurance (QA) and Hazard Analysis and Critical Control Point (HACCP) professionals in evaluating regulatory requirements, contamination risk, and safe processing alternatives. Historical FSIS establishment-performance categories provide an additional early-warning component, while the framework also

considers food retention, nutritional preservation, and potential distribution to communities with elevated food-security needs.

By combining food-safety evidence, operational decision support, and food-security outcomes, PoultryRescue seeks to facilitate more informed and transparent decision-making while maintaining compliance with regulatory requirements and protecting public health. The application is not intended to determine product safety, authorize product release, or estimate the number of illnesses prevented. Rather, it provides an explainable framework that helps qualified professionals evaluate potential interventions and the trade-offs associated with each option. This integrated approach aligns with the One Health principles, recognizing the interconnected nature of food production, human health, environmental conditions, and equitable access to nutritious food as essential components of a resilient food system.

Problem Statement and Rationale

Foodborne illness continues to pose a substantial public health burden in the United States, with nontyphoidal *Salmonella* among the leading causes of foodborne infections. Poultry products, especially chicken, are recognized as important vehicles for *Salmonella* transmission because contamination can occur throughout production, processing, distribution, handling, and preparation. The Centers for Disease Control and Prevention (CDC, 2025b) estimates that approximately one million illnesses in the United States each year are associated with contaminated poultry. In addition, source-attribution studies conducted through the Interagency Food Safety Analytics Collaboration (IFSAC) have consistently identified chicken as a major contributor to foodborne *Salmonella* infections (CDC, 2026a). Together, these findings highlight the continuing need for effective preventive strategies that strengthen pathogen control across the poultry supply chain.

To evaluate and monitor pathogen-control performance, the United States Department of Agriculture Food Safety and Inspection Service (USDA-FSIS) established *Salmonella* performance standards and an establishment categorization system for poultry-processing facilities. Under this framework, establishments are assigned to one of three categories based on their performance relative to the applicable product-specific standard. Category 1 reflects consistent process control, Category 2 reflects variable process control while remaining within the standard, and Category 3 reflects highly variable process control that exceeds the standard during the evaluation period (USDA, 2016). Although these categorizations provide meaningful information regarding historical establishment performance, they do not directly measure the safety of individual poultry lots and are not intended to serve as stand-alone determinants of lot-disposition decisions.

Analysis performed by the Early Signal project team using the September 2026 FSIS establishment-category dataset revealed substantial differences in *Salmonella* performance among poultry product categories. Among 37 ground chicken processing lines with published categorization results, 21 were classified as Category 3. By comparison, only 48 of 436 categorized chicken-parts processing lines were assigned to Category 3. The analysis also found that 29 of the 37 categorized ground chicken lines were operated by small or very small establishments. These observations suggest that ground poultry warrants particular attention as a high-priority product category and provide a practical justification for targeting Quality Assurance (QA) and Hazard Analysis and Critical Control Point (HACCP) personnel at small and very small poultry-processing establishments as the primary users of the proposed decision-support system.

A critical gap remains between identifying Salmonella contamination or elevated contamination risk and determining the most appropriate management response. Even when microbiological test

results are available, processors must consider product-specific regulatory requirements, testing limitations, HACCP controls, and available processing alternatives. PoultryRescue seeks to bridge this gap by providing an explainable framework for evaluating appropriate actions, including additional verification, product holding, and validated further processing. This approach prioritizes food safety while identifying opportunities to avoid unnecessary food loss. Literature review

Burden and attribution

Scallan and colleagues estimated that nontyphoidal Salmonella accounted for approximately 11% of illnesses caused by major foodborne pathogens in the United States and was a leading contributor to foodborne-related hospitalizations and deaths (Scallan et al., 2011). More recent estimates for 2019 suggest that nontyphoidal Salmonella was responsible for approximately 1.3 million domestically acquired illnesses and 12,500 hospitalizations annually (Walter et al., 2025). Using outbreak-associated data, the Interagency Food Safety Analytics Collaboration (IFSAC) attributed 19.1% of Salmonella illnesses to chicken, while acknowledging that outbreak-based attribution may not fully represent sources of sporadic infections (CDC, 2026a). Supporting the importance of poultry as a reservoir, a genomic source-attribution study estimated that more than one-third of human Salmonella isolates with unknown exposure histories were likely attributable to chicken sources (Rose et al., 2025).

Performance standards and their limits

FSIS established maximum allowable Salmonella positivity rates of 25.0% for comminuted chicken, 13.5% for comminuted turkey, and 15.4% for chicken parts. Under this framework, establishments are assigned to Category 1 when all completed 52-week sampling windows during the previous three months remain at or below one-half of the applicable maximum standard, whereas establishments are assigned to Category 3 if any completed sampling window exceeds the standard (USDA, 2016). Implementation of these performance standards was associated with a reduction of more than 75% in Salmonella contamination of chicken parts (Williams et al., 2022). Despite these improvements in contamination levels, corresponding declines in human illness have not been observed to the same extent. A quantitative microbial risk assessment suggested that public health risk may be concentrated in a relatively small proportion of products containing high levels of highly virulent Salmonella serotypes (Kim et al., 2024). Furthermore, pathogen prevalence alone is an imperfect indicator of illness reduction and may not fully capture changes in public health risk (Ebel & Williams, 2015). Although FSIS proposed a revised regulatory framework, the proposal was withdrawn in 2025, leaving establishment performance categories as the primary publicly available plant-level indicator of Salmonella control performance (Federal Register, 2025).

These considerations define the scope of the present work. PoultryRescue is designed to predict Salmonella positivity, which serves as a measurable indicator of process performance and contamination risk. The system does not attempt to predict human illnesses, disease burden, or outbreak occurrence.

Plant-level evidence

Previous research has examined factors associated with establishment performance under FSIS *Salmonella* standards. One study found that performance relative to the standard was associated with establishment characteristics, including noncompliance histories and, to a lesser extent, older production facilities (Muth et al., 2009). Using retail surveillance data, Mujahid and colleagues reported that chicken-parts establishments classified as Category 3 exhibited higher *Salmonella*

positivity rates than establishments in lower categories, although this relationship reflected contemporaneous performance rather than predictive performance in subsequent years (Mujahid et al., 2023). Similarly, Li and colleagues analyzed FSIS chicken-parts data from FY2023 through FY2025 and identified several performance patterns, including differences associated with establishment size (Li et al., 2026). In another line of research, Lipman demonstrated that *Salmonella* genomic clusters can persist for years and span multiple companies, suggesting the presence of contamination reservoirs and transmission pathways that extend beyond individual processing establishments (Lipman, 2026).

The predictive component of PoultryRescue estimates subsequent Salmonella positivity as an indicator of establishment-level process performance. These estimates provide contextual risk information rather than direct evidence of contamination in individual poultry lots.

Forecasting, season and weather

Temperature has been associated with salmonellosis incidence in Europe, with the strongest effects observed near the time of consumption (Kovats et al., 2004). In the United States, county-level *Salmonella* risk has also been linked to extreme weather events and social conditions (Weller et al., 2024). Forecasting studies in other settings have generally found that models incorporating seasonal terms perform better than alternative approaches (Hatta et al., 2026). However, a recent FSIS analysis reported that a strong seasonal pattern is now observed only in ground beef, while seasonality appears to be weak for chicken carcasses (Williams et al., 2026). Our findings are consistent with this observation: since June 2023, chicken carcass *Salmonella* positivity has remained approximately 5% during both halves of the year.

Controls, lots and diversion

Effective control of *Salmonella* in poultry-processing operations depends on the implementation of validated microbial reduction interventions, robust food-safety management systems, and scientifically informed product-disposition decisions. Federal regulations require applicable fully cooked poultry products to achieve a **7-log$_{10}$ reduction of *Salmonella*** or an equivalent lethality treatment (Cornell Law School, 2026a), a requirement supported by extensive research demonstrating the thermal inactivation of *Salmonella* in poultry products (Juneja et al., 2001). HACCP systems further support pathogen control by establishing critical limits and control measures for significant food-safety hazards (Cornell Law School, 2026b), while ready-to-eat poultry products must also incorporate appropriate controls for *Listeria monocytogenes* (Cornell Law School, 2026c). Regulatory oversight has expanded to include certain not-ready-to-eat poultry products, such as breaded stuffed chicken products, for which USDA-FSIS established a *Salmonella* adulteration threshold of 1 CFU/g in 2024 (Salmonella Not Ready-To-Eat Breaded Stuffed Chicken Products, 2024).

In addition to these regulatory requirements, previous research has explored lot-level risk assessment as a strategy for reducing potential consumer exposure to *Salmonella*. Modeling studies involving ground chicken (Oscar, 2019) and ground turkey (Sampedro et al., 2024) have examined the potential benefits of identifying and diverting lots with higher contamination levels, demonstrating possible reductions in predicted foodborne illness while also highlighting trade-offs related to product availability and processing efficiency. However, the relationship between *Salmonella* exposure and subsequent illness is not uniform across serotypes, with substantial differences in pathogenicity and illness risk reported among strains (Teunis, 2022). Consequently, risk-assessment approaches based solely on contamination prevalence may not fully capture public health risk. Collectively, these findings support the development of decision-support frameworks

that integrate microbial risk information with validated processing alternatives and food-preservation considerations. PoultryRescue extends this concept by evaluating and comparing alternative lot-management actions while explicitly distinguishing its relative exposure metric from a quantitative estimate of illness risk.

The product-specific focus of PoultryRescue is also important. Certain not-ready-to-eat breaded stuffed chicken products have raised distinct concerns about consumers mistaking them for ready-to-eat foods (Salmonella Not Ready-To-Eat Breaded Stuffed Chicken Products, 2024), reinforcing the need to account for product characteristics and preparation instructions alongside processing controls. The framework should be viewed as one component of a broader prevention strategy, alongside established HACCP and sanitation measures, consumer education, and preharvest interventions under investigation, including poultry vaccination and probiotic approaches (Neelawala et al., 2024). The latter approaches were not evaluated in the present model.

Food security

Recall data illustrate how food-safety incidents can contribute to substantial food waste. Among recalled products for which recovery information was available, 12% were not recovered at all, while the median recovery rate among the remaining recalls was only 29% (Gorton & Stasiewicz, 2017). At the same time, food donation remains an important mechanism for preserving safe food that might otherwise be discarded. Donors who provide food in good faith are afforded legal protections under federal law, and these protections were expanded in 2023 (Cornell Law School, 2026d; US Congress, 2023). To help identify areas with greater need, county-level indicators can be derived from resources such as Map the Meal Gap (Gundersen et al., 2014), the Social Vulnerability Index (SVI) (CDC, 2024b; Flanagan et al., 2011), and CDC PLACES, which incorporated a food-insecurity measure in its 2025 release (Greenlund et al., 2022).

The gap

Prior research has described FSIS establishment categories and developed models to predict human illness or retail-level *Salmonella* positivity. However, to our knowledge, no published study has evaluated how effectively a publicly reported FSIS establishment category predicts *Salmonella* positivity in subsequent FSIS samples collected from the same processing line across multiple product categories. Similarly, we identified no studies that assess alternative lot-management actions using both relative exposure measures and indicators of food distribution to high-need counties. It should be noted that this conclusion is based on a targeted rather than exhaustive review of the literature.

PoultryRescue integrates two complementary functions: early-warning information derived from historical FSIS establishment performance and product-specific decision support informed by available microbiological testing results. The framework evaluates possible management actions within applicable regulatory and food-safety constraints, while comparing modeled pathogen exposure, food retention, and potential food-security outcomes. Its innovation lies in connecting contamination information to transparent processing decisions rather than treating pathogen detection or prediction as the final outcome.

Data

The project uses the three core challenge datasets and several additional public datasets.

Table.1 Dataset and their role

Dataset	Coverage	Role
CDC BEAM (PulseNet-derived)	State, month, 2018 to Sept 2026	Human isolates for the forecast
CDC NORS	2014 to 2024	Outbreak context by serotype
FSIS raw poultry sampling	Plant, Oct 2013 to June 2026	Positivity, persistence, category tests
FSIS category postings	Windows ending 2022 to Aug 2026	Published category test, counts
FSIS MPI directory	Current plants	Plant ID crosswalk, plant size
NOAA nClimDiv, Storm Events	State, month, 2014 onward	Weather inputs tested, storm history
CDC/ATSDR SVI 2022	County	County need
USDA Food Access Atlas	County from tract, 2019	County need
CDC PLACES 2025	2,288 counties, 38 states and DC	County food insecurity, modeled
IRS exempt organizations	County from ZIP	Food banks and pantries, approximate
USDA cooking yields, FoodData Central	National	Servings and protein

Additional publicly available datasets were incorporated to address information gaps not covered by the three core challenge datasets. The FSIS MPI directory linked establishment categories to sampling records and plant size; CDC/ATSDR SVI, USDA food-access indicators, and CDC PLACES characterized potential community need; and USDA cooking-yield references and FoodData Central converted product weight into estimated servings and dietary protein. Approximate food-bank locations from IRS exempt-organization records supported exploration of possible distribution partners, while NOAA weather and storm information was evaluated in the surveillance forecasts. These additions were selected for their specific analytical or operational roles, not as substitutes for lot-level microbiological testing or validated processing decisions.

Product classification followed the conventions used within the FSIS sampling program. Chicken-parts program samples recorded as comminuted products were classified as chicken parts rather than ground poultry products. Connecticut was excluded from county-level analyses because changes to its county-equivalent geographic codes prevented direct alignment with other county-based datasets.

Historical FSIS establishment-category records were obtained from Internet Archive versions of publicly available FSIS files because automated access to archived records on the FSIS website was not feasible. All datasets used in the project were publicly available. To protect establishment confidentiality, project outputs report only aggregated results and summary statistics; no individual processing establishments are identified.

Methods

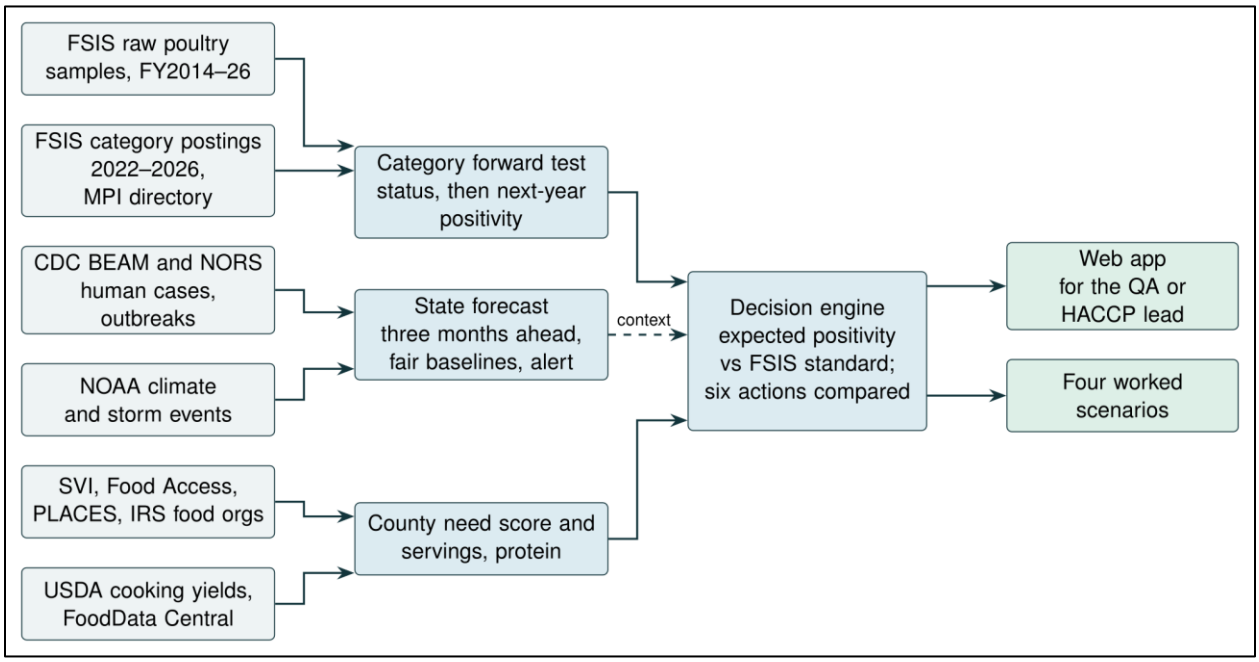


Fig.1. Data flow from source datasets through the PoultryRescue decision engine. State-level forecasts are provided as contextual information and do not influence risk classification.

Category forward test

Four archived FSIS establishment-category postings, corresponding to sampling windows ending in July 2022, 2023, 2024, and 2025, were used in the analysis. Published establishment categories were linked to subsequent FSIS routine sampling results through the MPI directory, with outcomes assessed during the 52-week period following each reporting window. Processing lines were required to have at least 10 samples within the evaluation period to be included. Confidence intervals were generated through plant-level resampling (Field & Welsh, 2007).

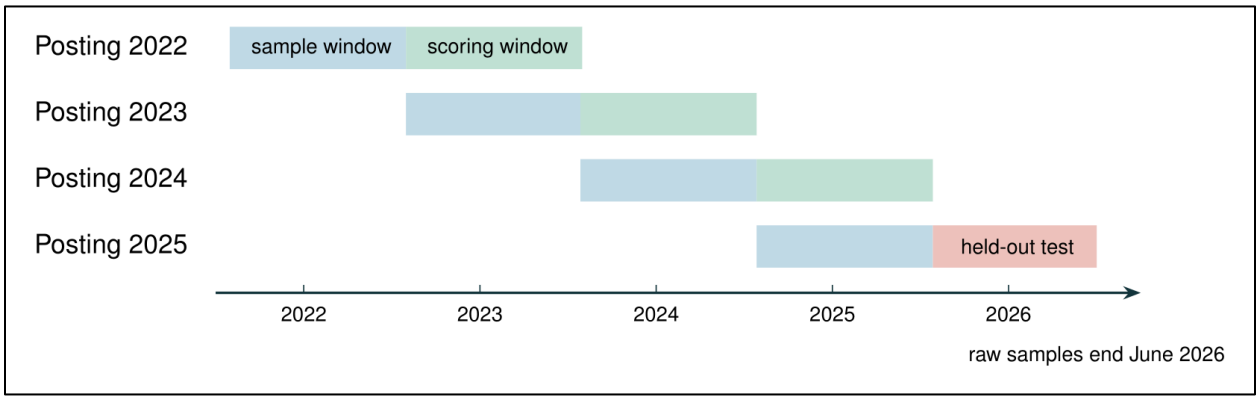


Fig.2. Forward-testing design is used to evaluate predictive performance. Categories were assigned using each FSIS posting's sampling window, and performance was assessed using subsequent FSIS samples collected after category publication. Dates are shown at the monthly level and are approximate.

We compared category plus product with product alone. Rates were set on the three earlier postings and scored on the last, using the Brier score, a proper scoring rule (Gneiting & Raftery, 2007). As a check, we rebuilt status from raw samples for windows starting September 2023, 2024 and 2025 and compared rebuilt status with published labels. Rebuilt status approximates the FSIS rule, which assesses several windows and uses count tables.

FSIS reported in December 2025 that samples from June and July 2025 had been missing, which affected postings from that period. The 2025 posting we used may predate that fix, so we repeated the test with it left out.

State case forecast

We fit a quasi-Poisson model by state and month on lagged counts, season and trend, with rolling-origin validation at a three-month horizon. Fair baselines were a same-month value, a three-year same-month mean and a seasonal model. A second model added FSIS, weather and storm inputs. An alert threshold was set on 2022-2023 and applied unchanged from 2024. All results use the latest data vintage. Serotype Kentucky was left out of the poultry serotype list, because a risk assessment found it contributes few cases (Kim et al., 2024).

Decision logic

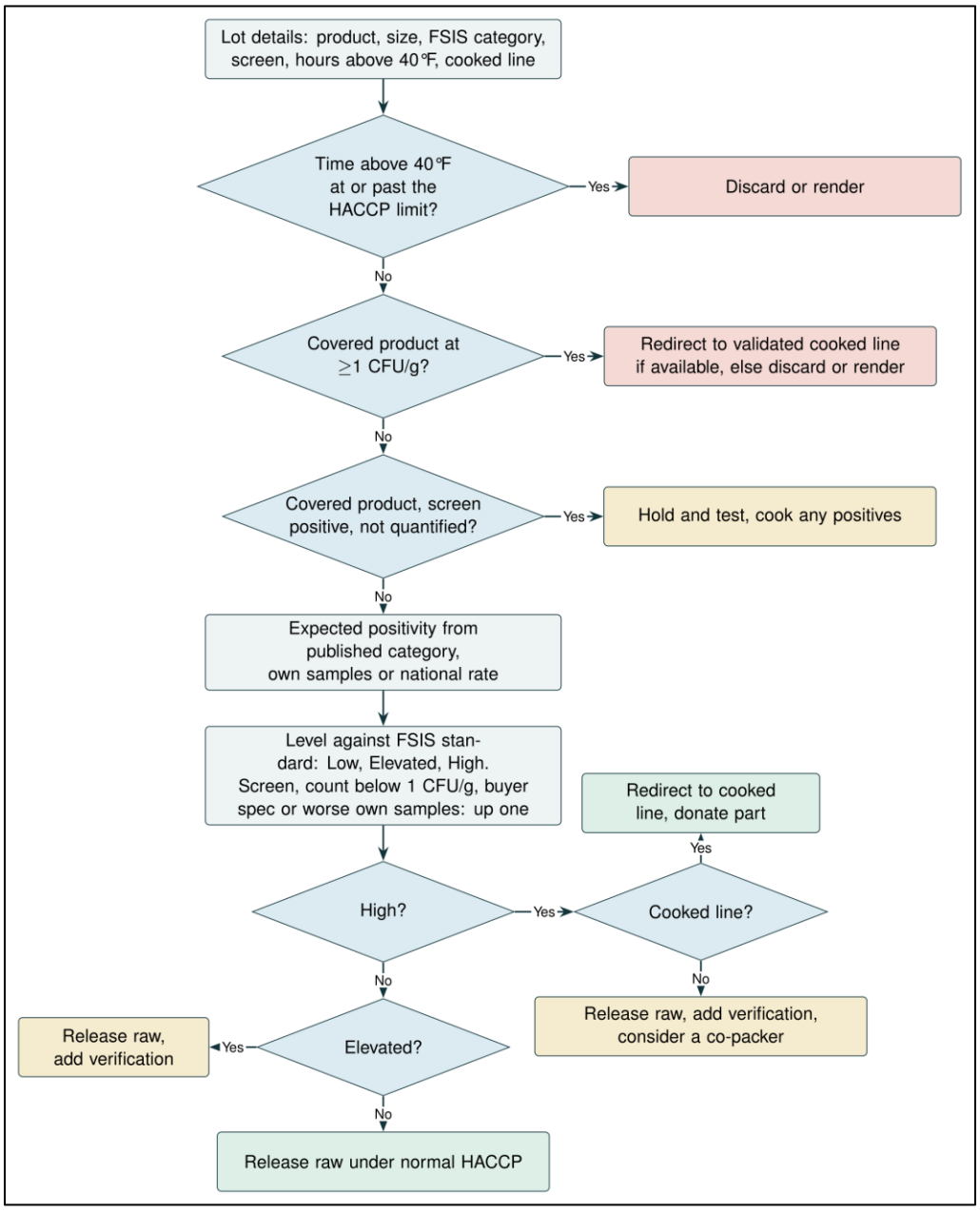


Fig.3. Decision logic.

Expected positivity is estimated using a hierarchical approach that first applies the published FSIS establishment category, then a category reconstructed from at least 10 establishment-specific samples, and finally the national product-level positivity rate when establishment-level information is unavailable. Risk levels are classified as Elevated when expected positivity exceeds one-half of the applicable FSIS performance standard and High when it exceeds the full standard. Because classification is based on point estimates, rates near a threshold may be assigned to the higher category; for example, ground chicken Category 1 is classified as Elevated. A positive screening result, a buyer specification requiring additional controls, or establishment-specific sampling results indicating poorer performance than the published category each increase the risk level by one category.

The decision framework distinguishes measured microbiological results from estimated contamination risk. Where applicable quantitative Salmonella measurements are available, product classification and relevant regulatory criteria inform the permitted management pathways. In the absence of such measurements, historical FSIS performance categories provide contextual estimates of expected positivity but do not establish contamination levels or determine the regulatory status of individual lots. Quantitative-testing pathways are currently illustrated through modeled scenarios and require further validation before operational implementation.
All management options are evaluated using a common set of outcome measures. The Exposure Index is calculated as expected positivity relative to the national chicken-parts positivity rate, multiplied by the proportion of product distributed in raw form. Estimated servings are based on 85-g cooked portions and USDA cooking-yield factors. Products diverted to a validated cooking process are assumed to eliminate exposure associated with raw product handling.

County need

The Need Score is calculated as the average percentile ranking of three indicators: the Social Vulnerability Index (SVI), the proportion of residents with low income and low access to food, and the CDC food-insecurity measure. In locations where a CDC PLACES food-insecurity estimate is unavailable, the score is based on the average of the first two indicators. Donated servings are allocated to the five highest-need counties within the establishment's state, weighted by the number of residents living below 150% of the federal poverty level.

Results

Published category predicts next-year positivity

Fig.4 shows the association between published USDA Food Safety and Inspection Service (FSIS) establishment categories and subsequent *Salmonella* positivity across four poultry product groups: ground chicken, ground turkey, chicken parts, and chicken carcasses. Across all product categories, next-year *Salmonella* positivity increased consistently from Category 1 to Category 3. Ground chicken exhibited the highest subsequent positivity rates, rising from 13.4% in Category 1 to 23.3% in Category 2 and 47.3% in Category 3. Ground turkey followed a similar pattern, with positivity increasing from 8.0% to 16.5% and 25.1%, while chicken parts increased from 6.3% to 10.0% and 17.5% across the three categories. Chicken carcasses showed the lowest positivity rates in Categories 1 (3.4%) and 2 (4.3%), but positivity increased markedly to 19.9% for Category 3 establishments.

For each poultry product group, Category 3 positivity exceeded the corresponding FSIS performance standard, as indicated by the horizontal dotted reference lines. Error bars represent 95% confidence intervals derived through plant-level resampling and reflect uncertainty in the estimated positivity rates. Collectively, these results indicate that published FSIS establishment categories are meaningfully associated with future *Salmonella* positivity and may serve as useful early indicators of elevated contamination risk. These findings support the inclusion of historical establishment performance within the PoultryRescue decision-support framework. However, the estimates reflect patterns observed in subsequent FSIS sampling results and should not be interpreted as direct predictions of contamination for individual poultry lots.

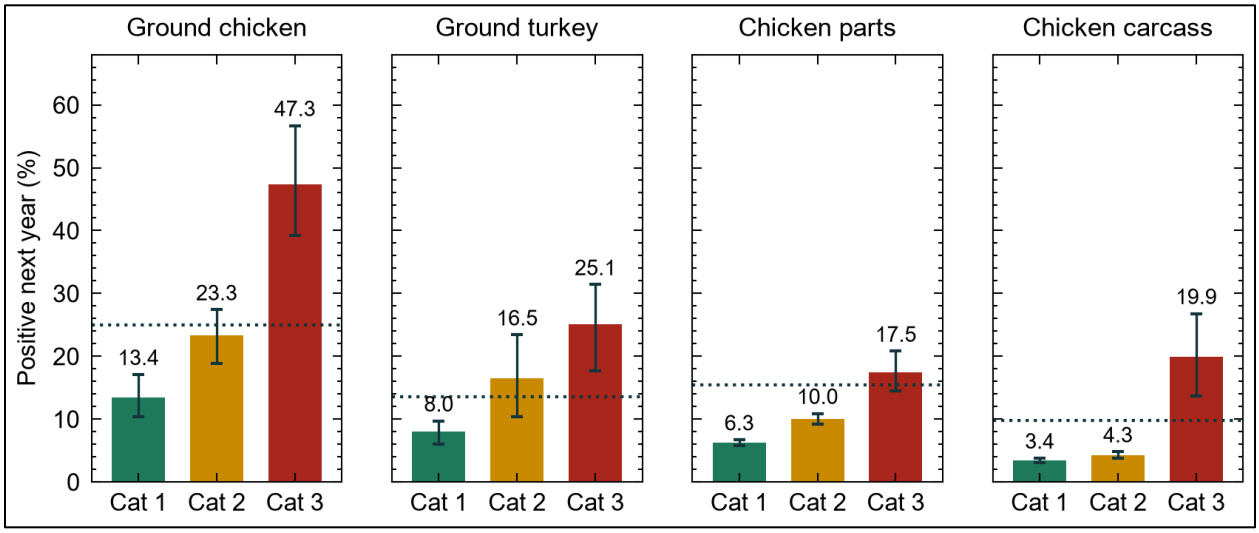


Fig.4. Next-year positivity after each published FSIS category. Whiskers show 95% intervals that resample plants, and dotted lines show the FSIS standard.

Table 2 provides the category-specific estimates and Category 3 confidence intervals across all four product groups. The largest absolute difference between Categories 1 and 3 was observed for ground chicken (33.9 percentage points). These are subsequent FSIS sampling outcomes at the processing-line level, not lot-specific probabilities of contamination.

Table 2. Next-year positivity by published category, four postings.

Product	Category 1	Category 2	Category 3 (95% CI)	FSIS standard
Ground chicken	13.4%	23.3%	47.3% (39.2 to 56.7)	25.0%
Ground turkey	8.0%	16.5%	25.1% (17.7 to 31.4)	13.5%
Chicken parts	6.3%	10.0%	17.5% (14.5 to 20.8)	15.4%
Chicken carcass	3.4%	4.3%	19.9% (13.7 to 26.8)	9.8%

Fig.5 presents year-to-year variation in subsequent *Salmonella* positivity among ground chicken and chicken parts processing lines by FSIS establishment category (Categories 1, 2, and 3) from 2023 through 2026. Across all reporting periods, Category 3 establishments consistently showed the highest subsequent positivity rates, followed by Category 2 and Category 1, indicating a stable relationship between historical establishment performance and future *Salmonella* positivity.

Among ground chicken processing lines, Category 3 positivity increased from approximately 41.6% in 2023 to 55.0% in 2026. In contrast, Category 1 positivity remained relatively stable, ranging from 11.8% to 15.1% over the study period. Category 2 positivity showed greater variability, rising to approximately 28% in 2025 before declining to approximately 21% in 2026. For chicken parts, positivity rates were comparatively stable across years, with Category 3 averaging approximately 17-19%, Category 2 approximately 9-11%, and Category 1 approximately 6-7%.

The sustained separation between categories across reporting periods suggests that published FSIS establishment classifications provide meaningful information about future contamination patterns.

The particularly high positivity rates observed among Category 3 ground chicken lines further support the selection of ground poultry as a priority focus for the PoultryRescue decision-support framework. However, estimates for 2026 should be interpreted with caution because they are based on a shorter observation period and may therefore be less directly comparable with results from previous years.

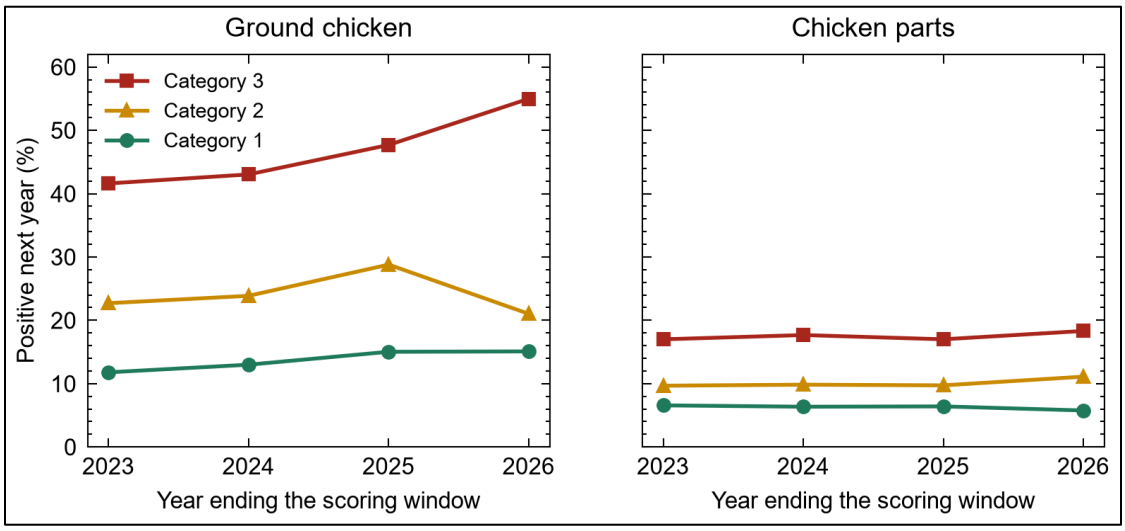


Fig. 5. The pattern held in each posting year. Ground chicken Category 1 ran 11.8% to 15.1%, and Category 3 ran 41.6% to 55.0%. The 2026 point rests on a shorter window.

State forecast

Predictive performance of five forecasting approaches was evaluated for three *Salmonella* surveillance outcomes: poultry-associated serotypes, *Salmonella* Enteritidis, and total *Salmonella* cases. Performance was measured using mean absolute error (MAE), defined as the average number of isolates per state-month over a three-month forecasting horizon, with lower values indicating greater predictive accuracy. Across all three outcomes, the history-based model (blue bars) achieved the lowest MAE and outperformed the three baseline approaches: the same-month model, the three-year historical mean, and the seasonal generalized linear model (GLM). These findings indicate that incorporating historical surveillance patterns improved forecasting accuracy relative to the baseline methods.

Expanding the history-based model to include additional USDA-FSIS sampling data and weather-related variables did not improve performance. Instead, the expanded model produced slightly higher MAE values across all outcomes, suggesting that these additional predictors provided limited incremental value within the forecasting framework. The results underscore the importance of empirical model validation and demonstrate that increasing model complexity does not necessarily lead to better predictions. Although environmental conditions and poultry contamination data may be epidemiologically relevant, their inclusion did not improve the accuracy of the three-month state-level forecasts evaluated in this study.

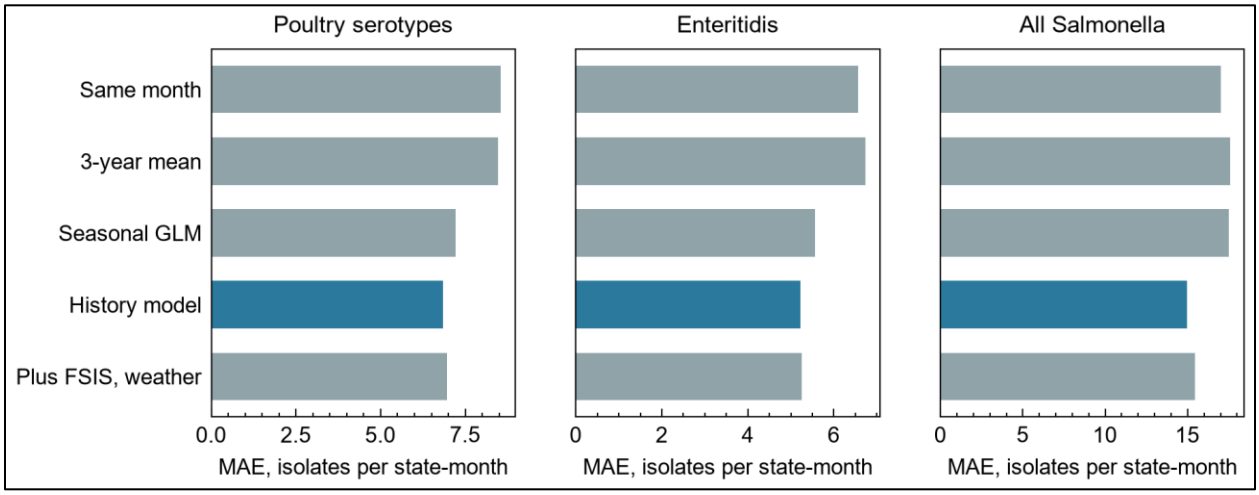


Fig.6. Three-month forecast error. The history model (blue) beat every fair baseline, and adding FSIS and weather data did not help.

The three-month forecasting analysis showed that the history-based model consistently outperformed the strongest baseline approach, reducing mean absolute error (MAE) by 5.1% for poultry-associated serotypes, 6.1% for *Salmonella* Enteritidis, and 12.1% for all *Salmonella* cases **Table.3**. Expanding the model to incorporate USDA-FSIS surveillance data, weather variables, and storm-related indicators did not produce additional improvements in predictive accuracy, suggesting that these variables contributed limited incremental value within the evaluated forecasting framework.

Evaluation of the associated risk-alert system yielded an overall precision of **53.8%** from 2024 onward, compared with a baseline event prevalence of **25.9%**. This indicates that alerts were more likely to coincide with elevated-risk periods than would be expected by chance alone. However, system recall was relatively low (**20.5%**), meaning that the majority of elevated-risk events were not identified. In addition, **73 of 158 alerts** were false positives, indicating a substantial number of warnings that did not correspond to observed elevated-risk events.

Table 3. Forecast and alert metrics.

Metric	Result
MAE against best fair baseline	5.1% lower (poultry serotypes), 6.1% (Enteritidis), 12.1% (all Salmonella)
Adding FSIS, weather and storms	No improvement
Alert precision, 2024 onward	53.8% against a 25.9% base rate
Alert recall	20.5%, with 73 false alarms in 158 alerts

Alert-system performance from 2024 through 2026 is shown by comparing alert precision with the underlying prevalence of elevated-risk events. Alert precision represents the proportion of issued alerts that correctly identified elevated-risk periods, whereas the base rate represents the overall frequency of elevated-risk events during each year.

The results indicate a steady decline in alert precision over time, from approximately 64.4% in 2024 to 49.1% in 2025 and 16.7% in 2026. While precision remained higher than the corresponding base rate in 2024 and 2025, it fell below the base rate in 2026, suggesting that the alert system no longer provided meaningful discrimination during that period. Interpretation of the 2026 estimate should be cautious because it was based on only 18 alerts, limiting the stability and

reliability of the annual performance estimate. The factors contributing to this decline have not yet been determined and warrant further investigation.

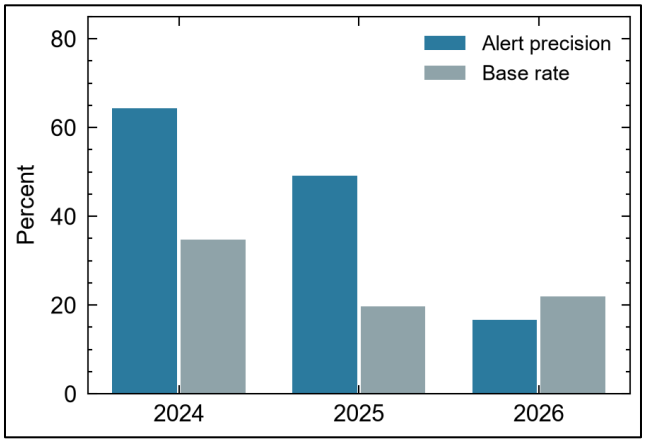


Fig. 7. Alert precision fell to 16.7% in 2026 on 18 alerts. We do not know why, so the alert does not affect the risk level.

Decision scenarios

Four illustrative poultry-processing scenarios were developed to evaluate the performance of the PoultryRescue decision-support framework under varying levels of *Salmonella* risk, processing capabilities, and operational constraints. For each scenario, the framework compares expected *Salmonella* positivity, assigned risk level, recommended management action, changes in the relative exposure index, and food-security outcomes, including servings retained and servings potentially directed to high-need communities.

In the first scenario **Table.4**, ground chicken originating from a Category 3 processing line had an expected *Salmonella* positivity of 47.3% and was classified as high risk. The recommended response was to divert to a validated cooking process, accompanied by donation of a portion of the finished product. Under model assumptions, this strategy reduced the relative raw-handling exposure index from 5.63 to 0.00 while preserving an estimated 38,422 servings, including 9,605 servings potentially distributed to high-need communities. By contrast, the second scenario involved chicken breasts from a Category 1 establishment with an expected positivity of 6.3%, resulting in a low-risk classification. The framework recommended release under standard HACCP controls, maintaining an exposure index of 0.75 while retaining approximately 38,422 servings.

The third scenario evaluated a breaded stuffed chicken product containing a measured *Salmonella* concentration of 2.0 CFU/g. This lot was classified as high risk, and diversion to a validated cooking process was recommended. The intervention reduced the modeled exposure index from 1.00 to 0.00 while preserving an estimated 19,211 servings. The fourth scenario involved a poultry thigh lot that experienced temperatures exceeding 40°F for six hours following a storm-related power outage. Because critical temperature control was compromised, the framework assigned a

high-risk classification and recommended disposal or rendering of the product, resulting in no retained servings.

Collectively, the scenarios demonstrate how PoultryRescue balances microbial-risk considerations with potential food-security impacts when evaluating alternative lot-management strategies. The examples illustrate that validated processing alternatives can preserve substantial quantities of nutritious food when appropriate controls are available, whereas events involving the loss of critical food-safety controls may necessitate product disposal despite the resulting food loss. The reported exposure indices and serving estimates are model-generated projections and should not be interpreted as direct measures of illness reduction or verified food recovery outcomes. All recommended actions remain subject to applicable regulatory requirements, validated processing procedures, and review by qualified food-safety personnel.

Table 4. Worked scenarios.

Scenario	Risk	Expected positivity	Recommended action	Exposure, raw to chosen	Servings kept / to need
Ground chicken, Category 3 line, cooked line available	High	47.3%	Redirect to cooked line, donate part	5.63 to 0.00	38,422 / 9,605
Chicken breast, Category 1 line, no cooked line	Low	6.3%	Release raw under normal HACCP	0.75 to 0.75	38,422 / 0
Breaded stuffed chicken lot measured at 2.0 CFU/g	High	8.4%	Redirect to validated cooked line	1.00 to 0.00	19,211 / 0
Thigh lot, product 6 hours above 40°F after a storm outage	High	6.3%	Discard or render	0.75 to 0.00	0 / 0

The comparison of six potential management actions for a high-risk ground chicken lot from a Category 3 processing line highlights the trade-offs between minimizing potential *Salmonella* exposure and preserving food for consumption **Fig.8.** The left panel compares the relative exposure index associated with each intervention, whereas the right panel shows the estimated number of servings retained and potentially distributed to high-need communities. Releasing the product as raw ground chicken or implementing additional verification measures resulted in the highest modeled exposure index, approximately 5.63, while holding and testing substantially lowered projected exposure. Redirecting the product to a validated cooking process reduced the modeled raw-handling exposure index to 0.00 while preserving an estimated 38,422 servings, demonstrating that exposure reduction and food retention can be achieved simultaneously under appropriate processing conditions.

The food-security outcomes further illustrate the value of incorporating nutritional preservation into food-safety decision-making. Under the modeled assumptions, diverting product to a validated cooking operation and donating 25% of the resulting output could provide approximately 9,605 servings to high-need communities while retaining the full estimated food volume. By comparison, discarding or rendering the product also reduced the modeled exposure index to 0.00, but resulted in the complete loss of potentially recoverable food. These results demonstrate how alternative

management strategies can differ substantially in their implications for both food safety and food security.

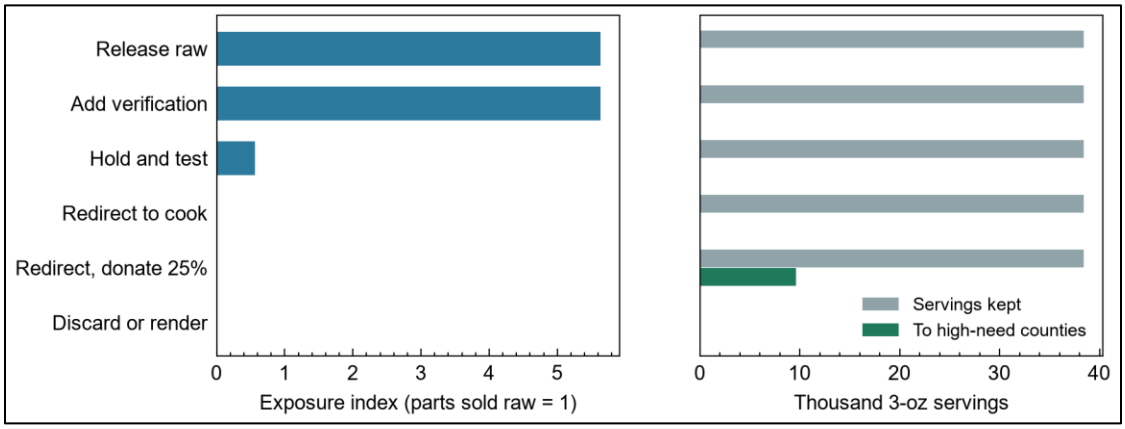


Fig. 8. Scenario 1 trade-off. Redirecting to a validated cook removes raw-handling exposure and keeps all servings, and a 25% donation sends 9,605 servings to high-need counties. Discarding also removes exposure, but it loses all food.

The scenario analysis also demonstrates the importance of considering operational feasibility and food-safety requirements when selecting management actions. Scenario 1 shows that food-safety and food-security objectives can be achieved simultaneously when access to validated cooking capacity is available. Under these conditions, additional processing can reduce modeled raw-handling exposure while preserving substantial quantities of nutritious food. For establishments that lack validated cooking facilities, the framework evaluates alternatives such as additional verification activities and, where appropriate, collaboration with qualified co-packers, provided that all applicable regulatory requirements are met.

In contrast, Scenario 4 highlights situations in which food-safety constraints limit opportunities for product recovery. Following a loss of temperature control, further processing cannot be assumed to eliminate all hazards associated with temperature abuse or reestablish compliance with an exceeded HACCP critical limit. Consequently, when the safety of additional processing cannot be adequately demonstrated, the framework gives priority to food-safety considerations over food-preservation objectives. These examples illustrate how PoultryRescue balances risk reduction, regulatory compliance, and food-security outcomes while recognizing that not all products can be safely redirected to alternative uses.

Food security

The relationship between county-level adult food insecurity and the Centers for Disease Control and Prevention/Agency for Toxic Substances and Disease Registry Social Vulnerability Index (CDC/ATSDR SVI) was evaluated across 2,288 U.S. counties. The analysis identified a strong positive association between the two measures, with a Spearman correlation coefficient of **0.877** (approximately 0.88), indicating that higher levels of social vulnerability are generally associated with higher levels of food insecurity. The scatterplot shows a clear upward trend, although variability in food insecurity becomes greater among counties with the highest SVI values. These results suggest that social vulnerability and food insecurity reflect closely related aspects of community disadvantage. As a result, although the CDC PLACES food-insecurity measure

provides a more direct indicator of nutritional need, its strong correlation with SVI indicates that the two metrics contribute largely overlapping rather than independent information to the composite county-level need score.

These findings have important implications for the equitable distribution of food within the PoultryRescue decision-support framework. By combining CDC PLACES food-insecurity estimates, CDC/ATSDR SVI data, and USDA food-access indicators, the system identifies counties with relatively greater food-security needs and provides information on potential food-distribution organizations. However, analysis revealed that the two highest-need counties in Missouri had no organizations identified in the Internal Revenue Service (IRS) database. This observation should not be interpreted as evidence that food-assistance resources are unavailable, as the database may not capture church-operated pantries, informal food networks, or other community-based organizations. Accordingly, local verification of recipient organizations, storage capacity, and distribution infrastructure remain necessary before food donations are recommended.

Although food donation activities are supported by established legal protections, successful implementation also requires maintenance of appropriate cold-chain conditions and compliance with applicable food-safety requirements, including *Listeria monocytogenes* control measures for eligible ready-to-eat products. Overall, the results highlight the value of incorporating food-security indicators into poultry-processing decision-making while recognizing both the substantial overlap among vulnerability measures and the practical limitations of currently available food-distribution datasets **Fig.9**.

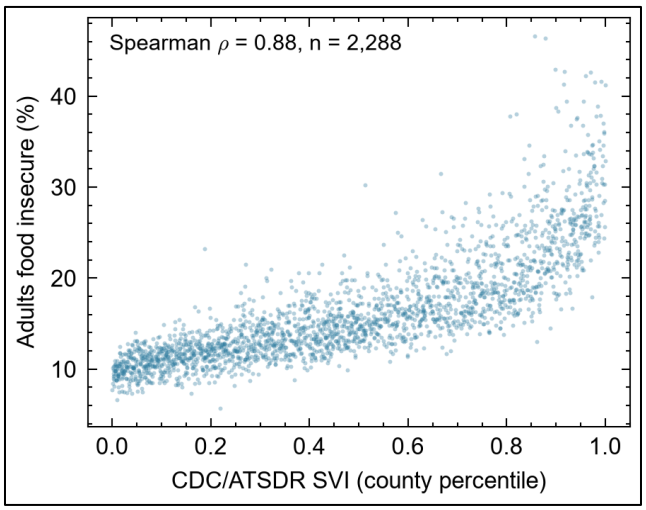


Fig.9. County food insecurity and SVI overlap strongly, so the need score should not count them as independent evidence.

The application

PoultryRescue provides an interactive interface for evaluating product-specific poultry-management decisions. Users can enter product characteristics, available microbiological testing information, and relevant operational constraints. The application displays a disposition recommendation and expected *Salmonella* positivity relative to applicable FSIS performance standards. Its decision logic identifies potential management pathways and compares their implications for food safety, processing, and nutritional preservation. Historical FSIS

establishment categories provide additional risk context and support early preparedness. Dedicated tabs explain risk classifications, compare management actions, and summarize potential food-security outcomes. The prototype demonstrates these functions through rule-based analysis and illustrative scenarios, including scenarios involving quantitative contamination measurements. The application is implemented as a static web tool and therefore does not require a server for operation. Its decision logic was validated against the underlying Python model across a range of test cases to ensure consistency. A separate Streamlit-based analyst interface contains mapping and forecasting components, while the primary application focuses on risk assessment, contextual surveillance alerts, scenario evaluation, management-action comparisons, and explanation of recommendations.

Kiro

The repo holds specs, steering files and a hook in .kiro/. The test hook and the plant self-check feature were built with Kiro's agent through a Quick Spec. Steering files, an earlier spec and the web app were made with Claude Code. At Waqar's request, Claude Code also operated the Kiro window for the Kiro work. .kiro/README.md records this and the manual changes.

Discussion and Public Health Implications

Taken together, the establishment-category findings, decision scenarios, and county-level indicators illustrate a potential bridge from surveillance information to operational decisions. These use cases are illustrative; prospective evaluation with processing establishments and community partners is still needed. The results point to a practical opportunity: use publicly available, historical establishment-performance data to prompt a transparent review of product-specific management options rather than treat those data as proof of lot safety. PoultryRescue brings together surveillance interpretation, processing feasibility, and estimated food-preservation outcomes, while retaining qualified food-safety review and applicable regulatory controls as prerequisites for any operational action.
For small and very small poultry plants, the proposed application could make it easier for QA and HACCP professionals to compare potential responses, document trade-offs, and identify when validated cooking or qualified co-processing may preserve food that would otherwise be lost. The worked scenarios illustrate possible servings and protein preservation, but do not establish real-world recovery, improved health outcomes, or prevented infections.

Public health implications extend beyond the processing facility. Safe refrigeration, prevention of cross-contamination, and cooking poultry to an internal temperature of at least 165°F (74°C) remain essential to reducing consumer exposure to Salmonella. For food-insecure households, access to reliable cold storage, appropriate preparation equipment, and food-safety information may shape whether donated poultry can be handled safely. Product-specific concerns are also important, particularly when not-ready-to-eat breaded stuffed poultry may be mistaken for fully cooked food (Salmonella Not Ready-To-Eat Breaded Stuffed Chicken Products, 2024). Upstream measures being studied, including poultry probiotics and vaccination (Neelawala et al., 2024), are complementary preventive approaches; PoultryRescue addresses downstream processing decisions, not those interventions. These considerations also reflect feedback from an external infectious-disease expert.

A central future application of PoultryRescue is the integration of validated quantitative *Salmonella* testing into product-specific processing decisions. As reliable measurements become available and applicable FSIS requirements are implemented, the platform could support processors in evaluating

permissible management options, including validated further processing of eligible materials. Future development should prioritize validation of quantitative-testing pathways, product-specific regulatory decision rules, and operational safeguards. This would extend PoultryRescue beyond surveillance-based preparedness toward comprehensive contamination-informed decision support.

Future evaluation should include structured input from plant QA/HACCP personnel, food-safety specialists, processors, and food-distribution partners; testing of real processing constraints and capacity; assessment of decision consistency; and validation of logistics, safe food handling, and community distribution feasibility. Any broader rollout would depend on validated controls and prospective evidence that the tool improves decisions without compromising food safety.

Limitations

Several limitations should be considered when interpreting the findings and applying the PoultryRescue framework.

- The framework predicts *Salmonella* positivity rather than human illness. Public health risk is influenced not only by contamination prevalence but also by contamination level and serotype virulence, factors that are not captured by presence-absence sampling data (Ebel & Williams, 2015; Kim et al., 2024).
- FSIS verification sampling is not designed as a random population survey. FSIS cautions that these data should not be used to identify outbreaks, establish outbreak linkages, or infer causal relationships between samples and illness events.
- The classification of ground chicken Category 1 as Elevated Risk is based on a narrow margin. The observed positivity rate was 13.4% from 18 establishments (95% CI: 10.4-17.1%), compared with an Elevated threshold of 12.5%. Because risk categories are assigned using point estimates, this classification should be viewed as borderline.
- The number of eligible ground poultry processing lines was limited, with only 32 Category 1 and 29 Category 2 ground chicken line-postings available for analysis. In addition, the number of published category listings declined in 2025.
- The four evaluation years are not fully independent because each scoring period substantially overlaps with the sampling window used to generate the subsequent FSIS establishment category.
- Several sources of data loss may affect representativeness. Processing lines associated with establishments missing from the current MPI directory could not be linked, lines with fewer than 10 samples in the scoring period were excluded, very low-volume operations are not represented in the routine FSIS sampling programs used, and available raw sample data extended only through June 2026.
- The 2025 establishment-category posting may have preceded an FSIS correction related to previously omitted samples. With that posting left out, category plus product still beat product alone (Brier score 3.1% lower, 95% CI 1.6 to 4.6), and ground chicken ran 12.9%, 24.3% and 43.9% by category.
- Establishment-specific historical positivity rates provided slightly better predictive performance than published establishment categories. Nevertheless, categories remain valuable because they are publicly available and comparable across establishments.
- Regression to the mean is likely to influence results. Because a Category 3 designation follows a period of unusually high positivity, subsequent positivity rates would be expected to move toward the establishment's long-term average. This effect was not formally

quantified. To address this limitation, the application reports observed next-year positivity rates rather than assuming persistence of extreme values.

- The exposure index is a relative measure that accounts for product type and establishment category but does not incorporate serotype-specific virulence or contamination dose (Teunis, 2022).
- Individual sample results should not be interpreted as direct indicators of lot-level contamination. A single negative sample does not establish the safety of a large production lot.
- Assumptions regarding testing effectiveness and the proportion of product available for donation are illustrative. Information on establishment-specific processing capacity, operational costs, and inventory levels is not publicly available.
- The two-hour threshold above 40°F referenced in the framework is derived from FSIS consumer guidance. In practice, establishment-specific HACCP critical limits govern operational decisions.
- Alert-system performance declined substantially in 2026, raising concerns regarding long-term stability. In addition, forecasting validation was retrospective and based on the most recent available data vintage.
- The composite need score combines multiple modeled indicators that are partially overlapping in scope. Furthermore, CDC food-insecurity estimates were unavailable for 11 states, and the IRS organization database does not include many food pantries and community-based food-distribution programs. As a result, need estimates and donation-network information should be viewed as screening tools rather than comprehensive assessments of local food-security resources.

Reproducibility

Repository: https://github.com/waqarsaleem320/early-signal. App: https://waqarsaleem320.github.io/early-signal/. A clean install on Python 3.11 earlier regenerated all results with python run_all.py in about 5 minutes, and a fresh clone passed all tests. The repo holds pinned requirements, a data dictionary, 56 tests, and scripts for the web data and report figures.

Use of AI tools

AI coding assistants, Claude Code and the Kiro agent, wrote and reviewed much of the code and analysis and drafted material for this report. Team members directed this work, checked the results against the source data, and reviewed and edited the final text.

Conclusion

The findings of this study demonstrate that publicly available USDA Food Safety and Inspection Service (FSIS) establishment categories can serve as useful forward-looking indicators of *Salmonella* contamination patterns in poultry processing. Across four reporting periods, published FSIS categories consistently distinguished subsequent *Salmonella* positivity rates, with Category

3 establishments exhibiting substantially higher positivity than those classified in Categories 1 and 2. For ground chicken, next-year positivity increased from 13.4% in Category 1 to 23.3% in Category 2 and 47.3% in Category 3. Incorporating establishment category information in addition to product type improved predictive performance, reducing the Brier score by 4.3% relative to a product-only baseline. Although the state-level forecasting model outperformed conventional baseline approaches, the inclusion of additional FSIS surveillance and weather-related variables did not produce further gains in forecasting accuracy. Collectively, these results support the use of establishment-level performance measures as the primary basis for operational decision support, while broader surveillance forecasts are more appropriately retained as contextual information.

PoultryRescue demonstrates the potential to connect Salmonella surveillance and available microbiological testing information with transparent, product-specific management decisions. Historical FSIS establishment categories provide useful early-warning information, while the decision-support framework illustrates how alternative processing actions can be evaluated in relation to regulatory requirements, modeled exposure, and food preservation. Future integration of validated quantitative testing could further strengthen its relevance to operational decision-making. By prioritizing food safety while identifying opportunities to preserve nutritious poultry, PoultryRescue offers a foundation for more informed and sustainable food-processing practices.

Responses to the challenge questions

1. Decision-maker and decision. The QA or HACCP lead at a small or very small ground poultry plant decides what to do with one lot: release under normal HACCP, release with added verification, hold and test, redirect to a validated cooked line (with or without donating part), or discard. The decision is made while the lot is under establishment control, using expected next-year positivity for the line.

2. Core datasets. Human signal: CDC BEAM (PulseNet-derived) isolates and NORS outbreaks. Food and animal signal: FSIS raw poultry sampling and published establishment categories. Environmental signal: NOAA nClimDiv climate and Storm Events. Food-security signal: CDC PLACES food insecurity, CDC/ATSDR SVI and the USDA Food Access Research Atlas.

3. Added datasets. Archived FSIS category postings and the MPI directory made the plant-level forward test possible. SVI, the Food Access Atlas, CDC PLACES and IRS food organizations target donations to high-need counties. USDA cooking yields and FoodData Central convert lots into servings and protein. Adding FSIS, weather and storm inputs did not improve the state forecast, and we report that result.

4. Modeled outcome. Next-year Salmonella positivity of a plant line given its published FSIS category and product. It is public, measured on later FSIS samples, and available before a lot decision is needed.

5. Performance. Category plus product beat product alone: Brier score 4.3% lower (95% CI 2.9 to 6.0) on 22,084 held-out samples, and 3.1% lower without the 2025 posting. The category pattern held in each of four posting years. The state forecast had 5.1% to 12.1% lower error than fair baselines, but alert precision fell from 64.4% in 2024 to 16.7% in 2026, so alerts are context only.

6. Errors that matter. Missed risk at a Category 3 line can expose consumers; false alarms waste food and money. The state alert found only 20.5% of surges and 73 of 158 alerts were false, so alerts never change the risk level, and every high-consequence decision requires review by a qualified person.

7. Conflicts between safety and food security. For high-risk lots, discarding removes exposure but loses all food. The decision logic prefers a validated cook with partial donation when a cooked line exists, and recommends discard only when HACCP control is lost, as in the storm-outage scenario.

8. Walk-through. A Category 3 ground chicken line has expected positivity of 47.3% (High). With a cooked line available, the tool recommends redirecting to a validated cook and donating 25%: the raw-handling exposure index falls from 5.63 to 0, 38,422 servings are kept and 9,605 go to high-need counties. The user sees the gauge with its 95% range, the reasons, the action comparison and the county list.

9. Team science. Nursing and food science input moved the project from a state forecast toward a plant-level decision tool, added food-security outcomes, the breaded stuffed chicken rule, and the Listeria and cold-chain conditions for donation. Engineering and computer science input shaped the validation design and the application.

10. Scope. Today the tool compares lot actions with stated uncertainty. It never claims a lot is safe, predicts outbreaks, counts illnesses avoided or makes a regulatory determination. Real-world use would need prospective validation with plant QA staff, plant capacity and cost data, and serotype and enumeration data.

References

CDC. (2024a, October 4). *About Salmonella Infection*. Salmonella Infection (Salmonellosis). https://www.cdc.gov/salmonella/about/index.html

CDC. (2024b, October 22). *Social Vulnerability Index*. Place and Health - Geospatial Research, Analysis, and Services Program (GRASP). https://www.atsdr.cdc.gov/place-health/php/svi/index.html

CDC. (2025a, September 24). *People at Increased Risk for Food Poisoning*. Food Safety. https://www.cdc.gov/food-safety/risk-factors/index.html

CDC. (2025b, November 25). *Chicken and Food Poisoning*. Food Safety. https://www.cdc.gov/food-safety/foods/chicken.html

CDC. (2026a, May 7). *IFSAC 2023 Annual Report*. Interagency Food Safety Analytics Collaboration. https://www.cdc.gov/ifsac/php/data-research/annual-report-2023.html

Choi, J., Kong, B., Bowker, B. C., Zhuang, H., & Kim, W. K. (2023). Nutritional Strategies to Improve Meat Quality and Composition in the Challenging Conditions of Broiler Production: A Review. *Animals*, *13*(8), 1386. https://doi.org/10.3390/ani13081386

Cornell Law School. (2026a). *9 CFR § 381.150—Requirements for the production of fully cooked poultry products and partially cooked poultry breakfast strips.* Legal Information Institute. https://www.law.cornell.edu/cfr/text/9/381.150

Cornell Law School. (2026b). *9 CFR § 417.2—Hazard Analysis and HACCP Plan.* Legal Information Institute. https://www.law.cornell.edu/cfr/text/9/417.2

Cornell Law School. (2026c). *9 CFR Part 430—REQUIREMENTS FOR SPECIFIC CLASSES OF PRODUCT*. Legal Information Institute. https://www.law.cornell.edu/cfr/text/9/part-430

Cornell Law School. (2026d). *42 U.S. Code § 1791—Bill Emerson Good Samaritan Food Donation Act*. Legal Information Institute. https://www.law.cornell.edu/uscode/text/42/1791

Ebel, E. D., & Williams, M. S. (2015). When Are Qualitative Testing Results Sufficient To Predict a Reduction in Illnesses in a Microbiological Food Safety Risk Assessment? *Journal of Food Protection*, *78*(8), 1451–1460. https://doi.org/10.4315/0362-028X.JFP-15-042

Federal Register. (2025). Salmonella framework for raw poultry products: Notice of withdrawal (90 FR 17344). https://www.govinfo.gov/content/pkg/FR-2025-04-25/html/2025-07187.htm

Field, C. A., & Welsh, A. H. (2007). Bootstrapping Clustered Data. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, *69*(3), 369–390. https://doi.org/10.1111/j.1467-9868.2007.00593.x

Flanagan, B. E., Gregory, E. W., Hallisey, E. J., Heitgerd, J. L., & Lewis, B. (2011). A Social Vulnerability Index for Disaster Management. *Journal of Homeland Security and Emergency Management*, *8*(1), Article 3. https://doi.org/10.2202/1547-7355.1792

FSIS. (2025, December 2). *Delayed Verification Sampling of Not Ready-to-Eat Breaded Stuffed Chicken Products*. Federal Register. https://www.federalregister.gov/documents/2025/12/02/2025-21737/delayed-verification-sampling-of-not-ready-to-eat-breaded-stuffed-chicken-products

Gneiting, T., & Raftery, A. E. (2007). Strictly Proper Scoring Rules, Prediction, and Estimation. *Journal of the American Statistical Association*, *102*(477), 359–378. https://doi.org/10.1198/016214506000001437

Gorton, A., & Stasiewicz, M. J. (2017). Twenty-Two Years of U.S. Meat and Poultry Product Recalls: Implications for Food Safety and Food Waste. *Journal of Food Protection*, *80*(4), 674–684. https://doi.org/10.4315/0362-028X.JFP-16-388

Greenlund, K. J., Lu, H., Wang, Y., Matthews, K. A., LeClercq, J. M., Lee, B., & Carlson, S. A. (2022). PLACES: Local Data for Better Health. *Preventing Chronic Disease*, *19*, 210459. https://doi.org/10.5888/pcd19.210459

Gundersen, C., Engelhard, E., & Waxman, E. (2014). *Map the Meal Gap*: Exploring Food Insecurity at the Local Level. *Applied Economic Perspectives and Policy*, *36*(3), 373–386. https://doi.org/10.1093/aepp/ppu018

Hatta, H. M., Musa, K. I., Hairon, S. M., Fuzi, N. M. H. M., Mokhter, N., Sulaiman, N., Ayasamy, A., Ismail, R., Hashim, R., & Ng, J. W. J. (2026). Temporal dynamics and forecasting of laboratory-confirmed human nontyphoidal *Salmonella* cases in Malaysia, 2018–2024: Implications for surveillance. *Osong Public Health and Research Perspectives*. https://doi.org/10.24171/j.phrp.2026.0293

Juneja, V. K., Eblen, B. S., & Ransom, G. M. (2001). Thermal Inactivation of *Salmonella spp.* in Chicken Broth, Beef, Pork, Turkey, and Chicken: Determination of D- and Z-values. *Journal of Food Science*, *66*(1), 146–152. https://doi.org/10.1111/j.1365-2621.2001.tb15597.x

Kim, M., Barnett-Neefs, C., Chavez, R. A., Kealey, E., Wiedmann, M., & Stasiewicz, M. J. (2024). Risk Assessment Predicts Most of the Salmonellosis Risk in Raw Chicken Parts is Concentrated in Those Few Products with High Levels of High-Virulence Serotypes of Salmonella. *Journal of Food Protection*, *87*(7), 100304. https://doi.org/10.1016/j.jfp.2024.100304

Kovats, R. S., Edwards, S. J., Hajat, S., Armstrong, B. G., Ebi, K. L., & Menne, B. (2004). The effect of temperature on food poisoning: A time-series analysis of salmonellosis in ten European countries. *Epidemiology and Infection*, *132*(3), 443–453. https://doi.org/10.1017/S0950268804001992

Li, Y., Lee, J., & Stasiewicz, M. J. (2026). Patterns and Trends in Salmonella-Positive, High-Level Salmonella, Serotypes, and Establishment Categorization in FSIS FY2023-25 Raw Chicken Parts

Laboratory Sampling Data. *Journal of Food Protection*, *89*(11), 100914. https://doi.org/10.1016/j.jfp.2026.100914

Lipman, D. J. (2026). Genomic epidemiology of *Salmonella* and *Campylobacter* in poultry production: Quantifying the contribution of primary breeders. *Proceedings of the National Academy of Sciences*, *123*(13), e2536554123. https://doi.org/10.1073/pnas.2536554123

Mujahid, S., Hansen, M., Miranda, R., Newsom-Stewart, K., & Rogers, J. E. (2023). Prevalence and Antibiotic Resistance of Salmonella and Campylobacter Isolates from Raw Chicken Breasts in Retail Markets in the United States and Comparison to Data from the Plant Level. *Life*, *13*(3), 642. https://doi.org/10.3390/life13030642

Muth, M. K., Fahimi, M., & Karns, S. A. (2009). Analysis of Salmonella Control Performance in U.S. Young Chicken Slaughter and Pork Slaughter Establishments. *Journal of Food Protection*, *72*(1), 6–13. https://doi.org/10.4315/0362-028X-72.1.6

Neelawala, R. N., Edison, L. K., & Kariyawasam, S. (2024). Pre-Harvest Non-Typhoidal Salmonella Control Strategies in Commercial Layer Chickens. Animals, 14(24), 3578. https://doi.org/10.3390/ani14243578

Oscar, T. P. (2019). Process risk model for *Salmonella* and ground chicken. *Journal of Applied Microbiology*, *127*(4), 1236–1245. https://doi.org/10.1111/jam.14395

Rabbitt, M. P., Reed-Jones, M., Hales, L. J., Suttles, S., & Burke, M. P. (2025). *Household food security in the United States in 2024*. Economic Research Service, U.S. Department of Agriculture. https://doi.org/10.32747/2025.9458834.ers

Rose, E. B., Steele, M. K., Tolar, B., Pettengill, J., Batz, M., Bazaco, M., Tameru, B., Cui, Z., Lindsey, R. L., Simmons, M., Chen, J., Posny, D., Carleton, H., & Bruce, B. B. (2025). Attribution of *Salmonella enterica* to Food Sources by Using Whole-Genome Sequencing Data. *Emerging Infectious Diseases*, *31*(4). https://doi.org/10.3201/eid3104.241172

Salmonella Not Ready-To-Eat Breaded Stuffed Chicken Products. (2024, May 1). Federal Register. https://www.federalregister.gov/documents/2024/05/01/2024-09393/salmonella-not-ready-to-eat-breaded-stuffed-chicken-products

Sampedro, F., Garcés-Vega, F., Strickland, A., & Hedberg, C. W. (2024). Developing a risk management framework to improve public health outcomes by enumerating and serotyping *Salmonella* in ground turkey. *Epidemiology and Infection*, 1–37. https://doi.org/10.1017/S0950268823002029

Scallan, E., Hoekstra, R. M., Angulo, F. J., Tauxe, R. V., Widdowson, M.-A., Roy, S. L., Jones, J. L., & Griffin, P. M. (2011). Foodborne Illness Acquired in the United States—Major Pathogens. *Emerging Infectious Diseases*, *17*(1), 7–15. https://doi.org/10.3201/eid1701.P11101

Teunis, P. F. M. (2022). Dose response for Salmonella Typhimurium and Enteritidis and other nontyphoid enteric salmonellae. *Epidemics*, *41*, 100653. https://doi.org/10.1016/j.epidem.2022.100653

US Congress. (2023). Food Donation Improvement Act of 2022 (Public Law 117-362). https://www.govinfo.gov/content/pkg/PLAW-117publ362/html/PLAW-117publ362.htm

USDA. (2016). *New Performance Standards for Salmonella and Campylobacter in Not-Ready-to-Eat Comminuted Chicken and Turkey Products and Raw Chicken Parts and Changes to Related Agency Verification Procedures: Response to Comments and* Announcement of Implementation Schedule. Federal Register, 81 FR 7285. https://www.govinfo.gov/content/pkg/FR-2016-02-11/html/2016-02586.htm

Walter, E. J. S., Cui, Z., Tierney, R., Griffin, P. M., Hoekstra, R. M., Payne, D. C., Rose, E. B., Devine, C., Namwase, A. S., Mirza, S. A., Kambhampati, A. K., Straily, A., & Bruce, B. B. (2025). Foodborne Illness Acquired in the United States—Major Pathogens, 2019. *Emerging Infectious Diseases*, *31*(4). https://doi.org/10.3201/eid3104.240913

Weller, D. L., Tierney, R., Verlander, S., Bruce, B. B., & Rose, E. B. (2024). Disparities in Salmonellosis Incidence for US Counties with Different Social Determinants of Health Profiles Are Also Mediated by Extreme Weather: A Counterfactual Analysis of Laboratory Enteric Disease Surveillance (LEDS) Data From 1997 through 2019. *Journal of Food Protection*, *87*(12), 100379. https://doi.org/10.1016/j.jfp.2024.100379

Williams, M. S., Bilanovic, I., Golden, N. J., & Venu, A. X. (2026). Seasonal Variation in the Occurrence of Salmonella in Raw Meat and Poultry Products in the United States from 2000 to 2024. *Journal of Food Protection*, 100953. https://doi.org/10.1016/j.jfp.2026.100953

Williams, M. S., Ebel, E. D., Golden, N. J., Saini, G., Nyirabahizi, E., & Clinch, N. (2022). Assessing the effectiveness of performance standards for Salmonella contamination of chicken parts. *International Journal of Food Microbiology*, *378*, 109801. https://doi.org/10.1016/j.ijfoodmicro.2022.109801